

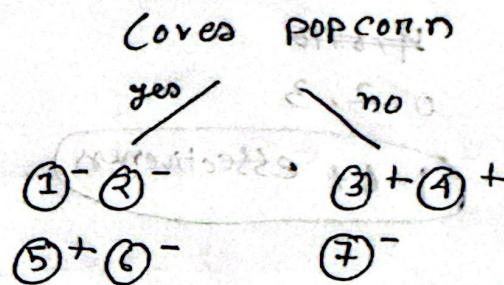
TOPIC NAME : Classification tree

DAY: _____

TIME: _____

DATE: / /

Loves Popcorn	Loves Soda	Age	Loves Troll 2
yes	yes	7	No
yes	no	12	no
no	yes	18	yes
no	yes	35	yes
yes	yes	38	yes
yes	no	50	no
no	no	83	no



$$\text{Gini (yes)} = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = 0.375$$

$$\text{Gini (no)} = 1 - \left(\frac{3}{3}\right)^2 - \left(\frac{1}{3}\right)^2 = 0.44$$

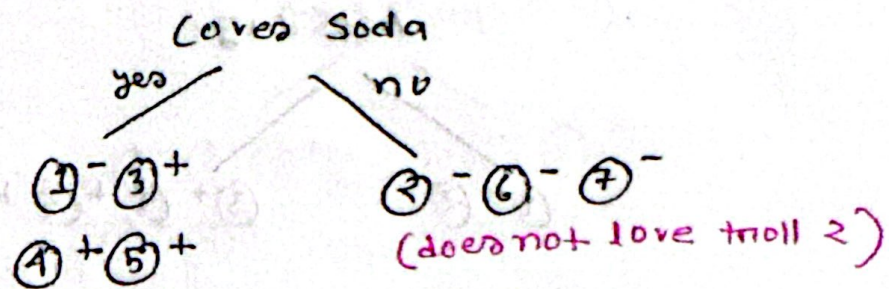
$$\begin{aligned} \text{Gini (Loves Popcorn)} &= \frac{4}{7} \times 0.375 + \frac{3}{7} \times 0.44 \\ &= 0.403 \end{aligned}$$

GOOD LUCK

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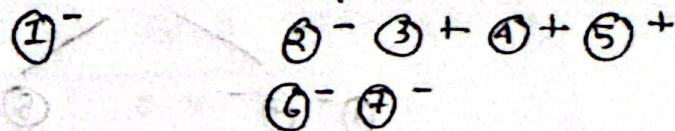
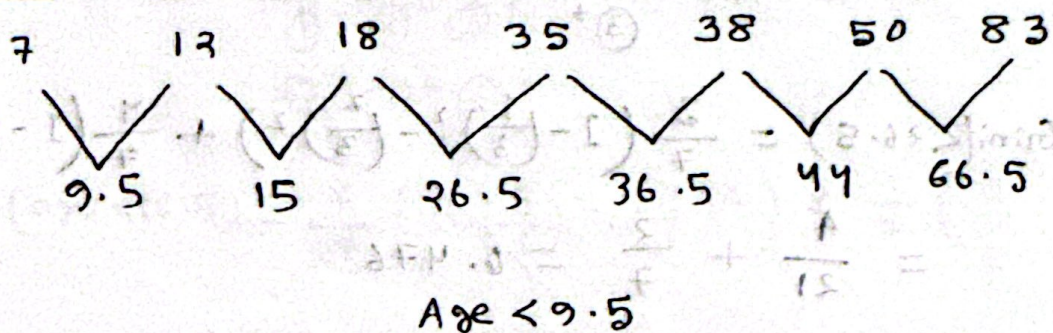


$$\text{Gini}(\text{yes}) = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

$$\text{Gini}(\text{no}) = 0$$

$$\text{Gini}(\text{Loves Soda}) = \frac{4}{7} \times 0.375 + 0 = 0.214$$

Age:



$$\text{Gini}(\text{Age} < 9.5) = \frac{1}{7} \left(1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2 \right) + \frac{6}{7} \left(1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 \right)$$

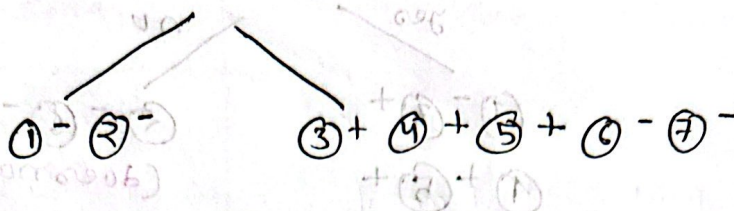
$$= 0.429$$

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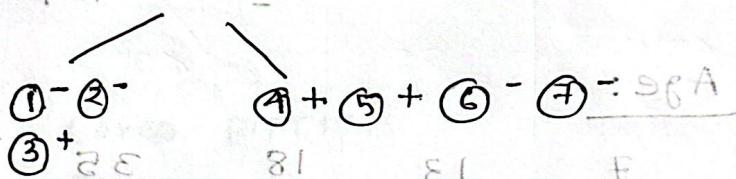
Age < 15



$$\text{Gini}(\text{age} < 15) = \frac{2}{7} \left(1 - \left(\frac{0}{2} \right)^2 - \left(\frac{2}{2} \right)^2 \right) + \frac{5}{7} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right)$$

$$= 0.343$$

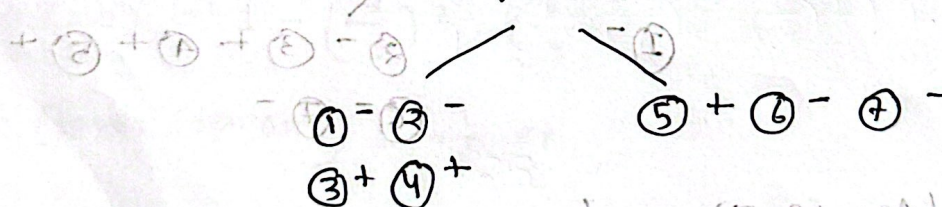
Age < 26.5



$$\text{Gini}(\text{age} < 26.5) = \frac{3}{7} \left(1 - \left(\frac{1}{3} \right)^2 - \left(\frac{2}{3} \right)^2 \right) + \frac{4}{7} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right)$$

$$= \frac{4}{21} + \frac{3}{7} = 0.476$$

Age < 36.5



$$\text{Gini}(\text{age} < 36.5) = \frac{4}{7} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right) + \frac{3}{7} \left(1 - \left(\frac{1}{3} \right)^2 - \left(\frac{2}{3} \right)^2 \right)$$

$$= \frac{2}{7} + \frac{1}{21} = 0.476$$

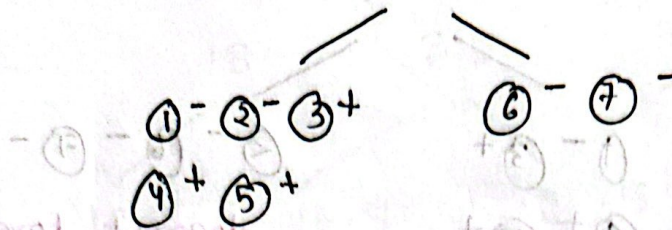
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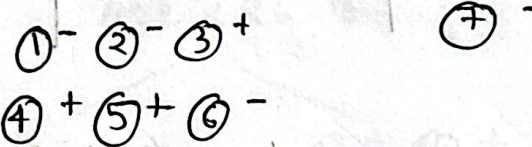
Age < 44



$$\text{Gini}(\text{age} < 44) = \frac{5}{7} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right) + \frac{2}{7} \left(1 - \left(\frac{0}{2} \right)^2 - \left(\frac{2}{2} \right)^2 \right)$$

$$= 0.343$$

Age < 66.5



$$\text{Gini}(\text{age} < 66.5) = \frac{6}{7} \left(1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 \right) + \frac{1}{7} \left(1 - \left(\frac{0}{1} \right)^2 - \left(\frac{1}{1} \right)^2 \right)$$

$$= 0.429$$

Threshold	Gini
9.5	0.429
15	0.343 ✓
26.5	0.476
36.5	0.476
44	0.343 ✓
66.5	0.429

Loves Soda

① - ③⁺
④⁺ ⑤⁺

② - ⑥ - ⑦ -

doesn't love troll 2

	Loves popcorn	Loves Soda	age	Loves troll 2
①	yes	yes	7	no
③	no	yes	18	yes
④	no	yes	35	yes
⑤	yes	yes	38	yes

Loves popcorn

① - ④⁺ esp. ②⁺ ③⁺

$$\text{Gini (loves popcorn)} = \frac{2}{4} \left(1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right) + \frac{2}{4} \left(1 - \left(\frac{2}{2} \right)^2 - \left(\frac{0}{2} \right)^2 \right)$$

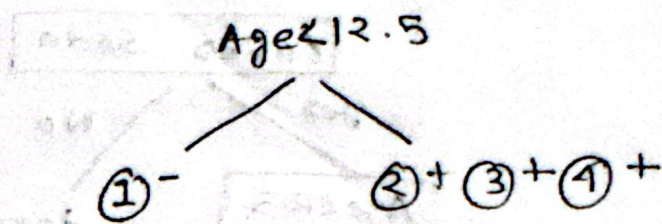
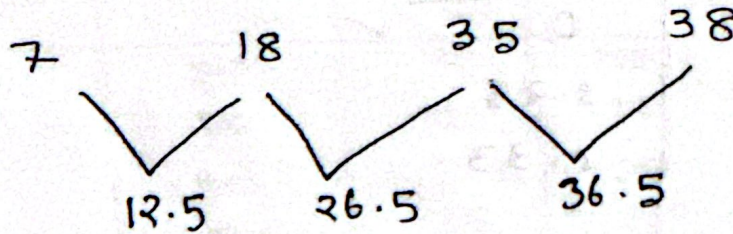
$$= 0.25 + 0.25 = 0.5$$

TOPIC NAME : _____

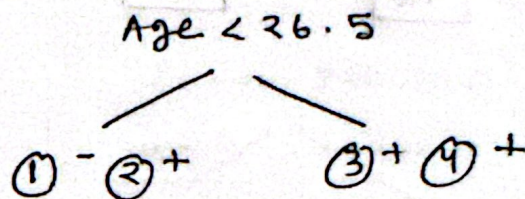
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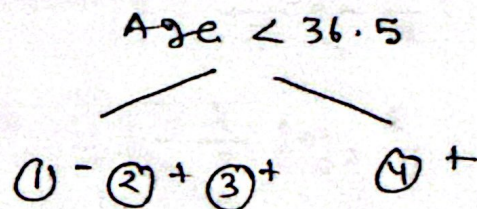
Age



$$\text{Gini}(\text{age} < 12.5) = 0$$



$$\begin{aligned} \text{Gini}(\text{age} < 26.5) &= \frac{3}{4} \left(1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \right) + 0 \\ &= 0.75 \end{aligned}$$



$$\begin{aligned} \text{Gini}(236.5) &= \frac{3}{4} \left(1 - \left(\frac{3}{3}\right)^2 - \left(\frac{1}{3}\right)^2 \right) + 0 \\ &= 0.33 \end{aligned}$$

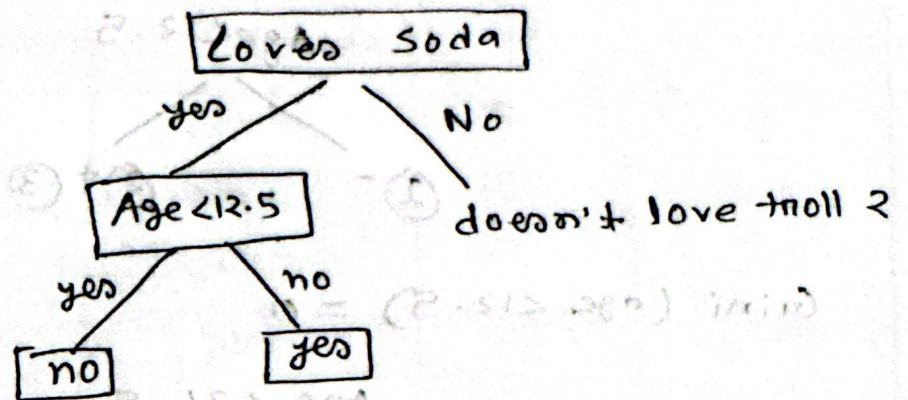
TOPIC NAME : _____

DAY : _____

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Threshold	Gini
12.5	0 ✓
26.5	0.25
36.5	0.33



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